MSG No. 192.247 $P6'/mcc'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `6'2'2`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,2,3,4,5,6,13,14,15,16,17,18]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[7,8,9,10,11,12,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `6'/m..`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,2,3,7,8,9,13,14,15,19,20,21]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[4,5,6,10,11,12,16,17,18,22,23,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[7,8,9,10,11,12]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[13,14,15,16,17,18]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[19,20,21,22,23,24]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[4,5,6,10,11,12]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[13,14,15,19,20,21]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[16,17,18,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `6'..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3,13,14,15]`
2	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	`[4,5,6,16,17,18]`
3	$[0,\, 0,\, -z]$	`[7,8,9,19,20,21]`
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	`[10,11,12,22,23,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, \frac{1}{4}\right]$	`[1,5,14,16]`
2	$\left[0,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[2,6,15,17]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$	`[3,4,13,18]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[7,12,21,22]
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[8,10,19,23]
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[9,11,20,24]

Table 7: Wyckoff site: 6g, site symmetry: 2'/m..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,8,14,19]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,9,15,20]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,7,13,21]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,12,18,22]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,10,16,23]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,11,17,24]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,2,3]
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[4,5,6]
3	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[7,8,9]
4	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	[10,11,12]
5	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[13,14,15]
6	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	[16,17,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[19,20,21]
8	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,14]
2	$\left[0, \frac{1}{2}, z\right]$	[2,15]
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[3,13]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[4,18]
5	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[5,16]
6	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[6,17]
7	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[7,21]
8	$\left[\frac{1}{2}, 0, -z\right]$	[8,19]
9	$\left[0, \frac{1}{2}, -z\right]$	[9,20]
10	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[10,23]
11	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[11,24]
12	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[12,22]

Table 10: Wyckoff site: `12j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,16]`
2	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,17]`
3	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,18]`
4	$\left[x,\ x,\ \frac{1}{4}\right]$	`[4,13]`
5	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,15]`
7	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[7,22]`
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[8,23]`
9	$\left[0,\ x,\ \frac{3}{4}\right]$	`[9,24]`
10	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[10,19]`
11	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[11,20]`
12	$\left[x,\ x,\ \frac{3}{4}\right]$	`[12,21]`

Table 11: Wyckoff site: `12k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[2,6]`
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,4]`
4	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[7,12]`
5	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[8,10]`
6	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[9,11]`
7	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[13,18]`
8	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[14,16]`
9	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[15,17]`
10	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[19,23]`
11	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[20,24]`
12	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[21,22]`

Table 12: Wyckoff site: `12l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$[-y,\ x-y,\ 0]$	`[2,9]`
3	$[-x+y,\ -x,\ 0]$	`[3,7]`
4	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[4,12]`
5	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[5,10]`
6	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[6,11]`
7	$[x-y,\ x,\ 0]$	`[13,21]`
8	$[-x,\ -y,\ 0]$	`[14,19]`
9	$[y,\ -x+y,\ 0]$	`[15,20]`
10	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[16,23]`
11	$\left[y,\ x,\ \frac{1}{2}\right]$	`[17,24]`
12	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[18,22]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	`[4]`
5	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[5]`
6	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[6]`
7	$[-x+y,\, -x,\, -z]$	`[7]`
8	$[x,\, y,\, -z]$	`[8]`
9	$[-y,\, x-y,\, -z]$	`[9]`
10	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[12]`
13	$[x-y,\, x,\, z]$	`[13]`
14	$[-x,\, -y,\, z]$	`[14]`
15	$[y,\, -x+y,\, z]$	`[15]`
16	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	`[16]`
17	$\left[y,\, x,\, \frac{1}{2}-z\right]$	`[17]`
18	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	`[18]`
19	$[-x,\, -y,\, -z]$	`[19]`
20	$[y,\, -x+y,\, -z]$	`[20]`
21	$[x-y,\, x,\, -z]$	`[21]`
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	`[22]`
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	`[23]`
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[24]`